real-data scaling laws. The scenario we study is highly inspired by these pioneering works, but still, in our view, different. We consider the case when we keep *augmenting* synthetic data (generated by the previous model trained on all the previous data so far) as iterations progress, much akin to how – in our view – the internet evolves. We observe that we can avoid model collapse in this setting. The analysis of previous models in our case is more involved, since the data used for training at iteration n is not homogeneous – different models from the past impart different statistical aspects to different parts of the training data. We also note a related augmentation model studied by Jain et al. (2024) – they perform risk minimization augmenting real data with synthetic data available from a potentially different independent source. One of their messages is that augmentation of (even) pure noise can be looked upon as adding a ridge penalty and hence, in certain cases, can improve test error. Their setup, however, is different from ours, since the synthetic data in their setup is not obtained by a learning algorithm employed on the real data, and the process is not iterative. However, morally, each iteration of ours involves risk minimization on data statistically composed of an equal mixture of data generated from the previous models, and hence each iteration of ours can be mapped to the general framework developed in Jain et al. (2024), although the dependencies among the various models trained in our setup introduce theoretical complications that do not seem to be too easily addressed by the theory developed in Jain et al. (2024). Shortly after v1 of our manuscript was uploaded to ArXiv, two other manuscripts appeared, dealing with the theoretical aspects in a setting similar to ours. Theorem 1 of Marchi et al. (2024) obtains the same square summability scaling of the variance as us. Seddik et al. (2024) studies collapse in language models in both purely synthetic and partly synthetic regimes and obtains deviation bounds as model iterations progress.

Considering Accumulating Data The two papers we found that partially considered accumulating data are Martínez et al. (2023a) and Alemohammad et al. (2023). Alemohammad et al. (2023) did so in one-half of one experiment: StyleGAN2 trained on FliqrFaces 128×128 (App. Fig. 8). The authors concluded that accumulating data does not avoid model collapse, but merely slows it down. However, we believe that a closer examination of their results (App. Fig. 8) reveals that accumulating data causes the test error to plateau to a relatively low error with increasing numbers of model-fitting iterations. This result would support our conclusion that accumulating data avoids model collapse and does not merely delay it. The results from Martínez et al. (2023a) are harder to evaluate; model collapse only seems to occur when the amount of synthetic data added per model-fitting iteration is 2× the total amount of accumulated data, and the subsequent work by the authors switched from

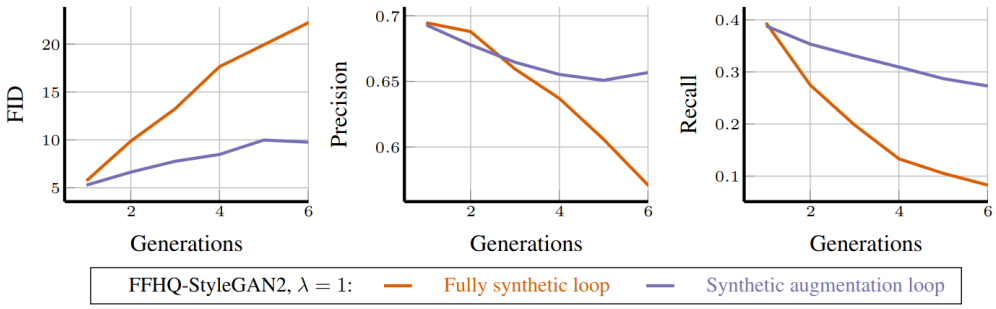

Figure 8: **Clarification of Data Accumulation in Alemohammad et al. (2023)**. Figure 7 from Alemohammad et al. (2023) (above) shows that linearly accumulating data ("Synthetic augmentation loop") causes poor behavior to plateau with the number of model-fitting iterations. Alemohammad et al. (2023) write, "Our experiments [...] support our main conclusion [that] fixed real training data only delays the inevitable degradation of the quality or diversity of the generative models over generations." We believe is that our evidence and their evidence is more consistent with the conclusion that accumulating data *avoids* model collapse and does not merely delay it.

accumulating data to replacing data (Martínez et al., 2023b). We think understanding what conditions and why these discrepancies exist is an interesting future direction.

Avoiding Model Collapse Several papers present methods for avoiding or slowing model collapse. Bertrand et al. (2023) shows in the replacing data setting that model collapse will not occur if the initial generative models approximate the data distribution well enough and the proportion of real data is sufficiently large with respect to the synthetic data. Dohmatob et al. (2024b) similarly demonstrates that in the replacing data setting, carefully selecting real data to mix with synthetic data can avoid model collapse. Other solutions may also be possible in various models and under various assumptions. To our knowledge, no paper has claimed an "optimal" strategy to avoid model collapse, and neither has ours.

B Proofs of Mathematical Results

We point out a lemma useful to prove Theorem 2.

Lemma 3. *Let T and d be positive integers with $T \geq d+2$, and let $X \in \mathbb{R}^{T \times d}$ be a random matrix with i.i.d. rows from $\mathcal{N}(0, \Sigma)$ with Σ positive definite. Then, X has full rank a.s. Moreover, it holds that:*

$$\mathbb{E}_X[(X^\top X)^{-1}] = \frac{1}{T-d-1}\Sigma^{-1}. \tag{5}$$

Proof. See Dohmatob et al. (2024a). □

Assuming Lemma 3 and Theorem 1, we present the proof of Theorem 2.

Proof of Theorem 2. From Theorem 1, we have:

$$\hat{w}_n = w^* + (X^\top X)^{-1} X^\top \left(\sum_{i=1}^{n} \frac{E_i}{i} \right) \tag{6}$$

where w^* is the true parameter, X is the original data matrix, and E_i are the noise terms at each iteration, with $E_i \sim \mathcal{N}(0, \sigma^2 I_T)$. The test error is given by:

$$E_{\text{test}}(\hat{w}_n) = \mathbb{E}[||\hat{w}_n - w^*||_\Sigma^2] \tag{7}$$

where the expectation is taken over all random quantities involved.

Substituting $\hat{w}_n$ into the test error expression and using the fact that $\Sigma \stackrel{\text{def}}{=} I_d$, we get:

$$\begin{aligned} E_{\text{test}}(\hat{w}_n) &= \mathbb{E}\left[\left(\sum_{i=1}^{n} \frac{E_i}{i} \right)^\top X(X^\top X)^{-2} X^\top \left(\sum_{i=1}^{n} \frac{E_i}{i} \right) \right] \\ &= \mathbb{E}\left[\sum_{i=1}^{n} \frac{\sigma^2}{i^2} \text{tr}(X(X^\top X)^{-2} X^\top) \right] \\ &= \sum_{i=1}^{n} \frac{\sigma^2}{i^2} \mathbb{E}\left[\text{tr}((X^\top X)^{-1}) \right] \end{aligned}$$

Using Lemma 3, we have:

$$\mathbb{E}_X\left[\text{tr}((X^\top X)^{-1}) \right] = \frac{d}{T-d-1} \tag{8}$$

Therefore, the test error for ridgeless regression with isotropic features in the data accumulation setting is:

$$E_{\text{test}}(\hat{w}_n) = \sum_{i=1}^{n} \frac{\sigma^2}{i^2} \cdot \frac{d}{T-d-1} < \frac{\sigma^2 d}{T-d-1} \left(\frac{\pi^2}{6} \right)$$

as $\sum_{i=1}^{n} i^{-2} < \sum_{i=1}^{\infty} i^{-2} = \pi^2/6$. □

Finally, we prove Theorem 1.

Proof of Theorem 1. We prove this theorem by induction.

Base case: For $n = 1$, we have:

$$\hat{w}_1 = \tilde{X}_1^\dagger \tilde{Y}_1 = (X^\top X)^{-1} X^\top (Xw^* + E_1) = w^* + (X^\top X)^{-1} X^\top E_1$$

which satisfies the lemma.